

**Final Report: Integrating Neural Networks and Machine Learning for PCOS Diagnosis
from Imaging and Clinical Data**

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Introduction	4
Background and Business Problem	4
Objectives and Research Questions	4
Contribution and Scope	4
Method	4
CRISP-DM Framework	4
Data Understanding	5
Clinical Tabular Dataset	5
Data Preparation	5
Tabular Data Preparation	5
Image Data Preparation	7
Models	8
Clinical Tabular Data - Classical Machine Learning Model	8
Clinical Tabular Data - Neural Network	9
Convolutional Neural Networks for Ultrasound Images	9
Classical Machine Learning Approach - Feature Extraction for Unstructured Data	12
Rationale for HOG in Ultrasound Imaging:	12
Implementation Details:	13
Dimensionality Reduction and SVM Classification:	13
Hyperparameter Tuning and Training Procedure	14
Hyperparameter Tuning Clinical Tabular Data	14
Hyperparameter Tuning Ultrasound Images	15
Evaluation Metrics	15
Tabular Dataset Results	15
Classical ML + Neural Network Performanc	15
.....	18
Ultrasound Image Dataset Results	18
CNN Performance	18
Classical ML on Extracted Features (Without Neural Networks)	20
Best Models and Confusion Matrices / ROC Curves:	22
Discussion	24

Results and Implications	24
Current Research Landscape	26
Business Use Analysis	29
Next Steps	29
References.....	31

Introduction

Background and Business Problem

Polycystic Ovary Syndrome (PCOS) is a chronic, multifactorial endocrine disorder and one of the leading causes of anovulatory infertility in women of reproductive age. Clinically, PCOS is characterized by a constellation of findings that may include hyperandrogenism, ovulatory dysfunction, metabolic disturbance, and polycystic ovarian morphology on ultrasound. In addition to reproductive consequences, PCOS is associated with long-term cardiometabolic risks, including insulin resistance, dyslipidemia, impaired glucose tolerance, and an elevated likelihood of developing type 2 diabetes and cardiovascular disease. Consequently, the timely identification of women with PCOS is not merely a diagnostic exercise but a critical entry point for preventive care and longitudinal risk management.

Objectives and Research Questions

The primary objective of this study is to develop and evaluate automated classification models for PCOS detection using ultrasound images, in the context of a broader dataset that also contains structured clinical information. Formally, the central prediction task is framed as a binary classification problem, in which each case is labeled as either PCOS (infected) or Non-PCOS (non-infected). Within this overarching objective, the work is structured around the following research questions:

1. Predictive performance of image-based models.

To what extent can a custom Convolutional Neural Network (CNN), trained end-to-end on ultrasound images, accurately differentiate PCOS from Non-PCOS cases? How does the CNN's performance compare to that of a classical computer vision pipeline based on Histogram of Oriented Gradients (HOG) descriptors followed by a Support Vector Machine (SVM) classifier?

2. Impact of feature representation and modeling paradigm.

How effective are hand-crafted image features (HOG) relative to automatically learned deep features (CNN) for the specific task of PCOS detection from ultrasound? Does explicit dimensionality reduction via Principal Component Analysis (PCA) combined

with SVM yield comparable or superior performance to a neural network trained directly on pixel intensities?

3. Implications for multimodal decision support.

Given that PCOS diagnosis in practice is informed by both clinical variables and imaging findings, what do the results of purely image-based models suggest about the potential value of combining tabular clinical data **and** ultrasound images into future multimodal systems? Which modeling paradigm classical machine learning or neural networks appears most suitable as the backbone for such integrated, real-world clinical decision support tools?

By addressing these questions, the study aims not only to report classification metrics, but also to elucidate the strengths and limitations of different modeling strategies in a manner that is directly relevant to clinical deployment and health-system decision-making.

Contribution and Scope

This study makes three main contributions to automated PCOS detection from ultrasound images. First, it provides a systematic comparison between a custom Convolutional Neural Network (CNN) and a well-tuned HOG + SVM pipeline, showing that a classical feature-engineering approach can achieve performance that rivals—and in this dataset, exceeds—that of a compact deep network. Second, it situates image-based modeling within a multimodal data context: the work includes careful preparation and harmonization of structured clinical data (e.g., identifiers, missing values, feature naming), thereby aligning the dataset with how PCOS is actually assessed in practice and laying a foundation for future clinical–imaging fusion. Third, it offers a transparent, reproducible pipeline, documenting preprocessing, feature extraction, model architectures, and hyperparameter choices for both neural and classical methods, along with detailed evaluation metrics such as confusion matrices and per-class precision and recall.

The scope of this work is intentionally focused. The analysis is restricted to binary classification (PCOS vs. Non-PCOS) on a single integrated dataset, without external multi-center validation. Clinical tabular variables are prepared and aligned but are not yet fused with images

into a joint predictive model; quantitative results are reported for image-only models. Finally, the proposed systems are framed as components for screening, triage, and decision support, rather than as replacements for clinician judgment or formal diagnostic criteria.

Method

CRISP-DM Framework

- Business Understanding: Radiology/OB-GYN triage
- Data Understanding: Class balance, missingness, device variability (images), feature distributions (tabular).
- Data Preparation: Patient-level splits, augmentations, imputation/standardization, engineered features.
- Modeling: Classic ML (LR/RF/GBT) vs DL (ResNet/EfficientNet, MLP for tabular).
- Evaluation: Accuracy, Precision, Recall, F1, ROC-AUC, PR-AUC, Confusion Matrices, calibration; runtime/latency.
- Deployment: Threshold policy (recall-first)

Data Understanding

Clinical Tabular Dataset

The dataset is a tabular clinical dataset on women assessed for polycystic ovary syndrome (PCOS) and infertility, originally collected in 2020 from patients attending 10 different hospitals across Kerala, India, and released publicly on Kaggle by Prasoon Kottarathil. It comprises roughly 541–543 anonymized patient records, each described by around 40–45 physical and clinical parameters (e.g., anthropometric measures, menstrual history, hormonal assays, ultrasound findings), providing a structured feature space rather than image “resolution.” The primary outcome variable encodes PCOS diagnosis (PCOS vs non-PCOS), with some versions of the dataset also supporting prediction of infertility-related outcomes. In the commonly used version, 177 of 541 subjects are diagnosed with PCOS and 364 are not, yielding a moderately imbalanced class distribution of approximately 33% PCOS-positive vs 67% controls. All records are de-identified and focus solely on clinical and physical attributes, making

this dataset suitable for developing and evaluating machine-learning models for PCOS risk prediction and clinical decision support.

Ultrasound Image Dataset (source, size, classes, resolution, class balance)

The second dataset uses a retrospective ultrasound image dataset curated from ovarian ultrasound studies collected across three clinics between 2018 and 2022. The dataset comprises a total of 11,784 grayscale ultrasound images organized into two diagnostic classes: infected/PCOS-positive ovaries (6,784 images) and noninfected/healthy ovaries (5,000 images). This yields a moderately imbalanced class distribution of approximately 57.5% PCOS-positive and 42.5% healthy cases. Images originate from diverse acquisition settings, resulting in a multi-resolution collection ranging from 255×247 px to 984×848 px, thereby reflecting realistic variability in clinical imaging workflows. All patient identifiers have been removed and DICOM metadata stripped to ensure privacy, and images were annotated by radiology residents under consultant supervision.

Data Preparation

Tabular Data Preparation

Two PCOS tabular datasets were loaded and merged using “Patient File No.” as the key. Before merging, patient identifiers were standardized (format cleaned and made consistent) to ensure the join matched correctly. After the merge, the combined dataset shape was checked to confirm that the expected number of rows and columns carried through and to catch merge issues early (missing matches or accidental duplication).

To make the inputs clean and consistent and keep the evaluation fair across models, class imbalance was treated as a key issue because PCOS is the minority class (Figure 1), and high overall accuracy can still happen when the model performs poorly on PCOS cases. For the tabular clinical dataset, the target label was defined as a binary outcome with No PCOS = 0 and PCOS = 1, and this labeling was kept consistent across the notebook outputs and report tables. Non-informative fields (such as IDs or columns that do not represent meaningful patient features) were removed to reduce noise and prevent shortcut learning.

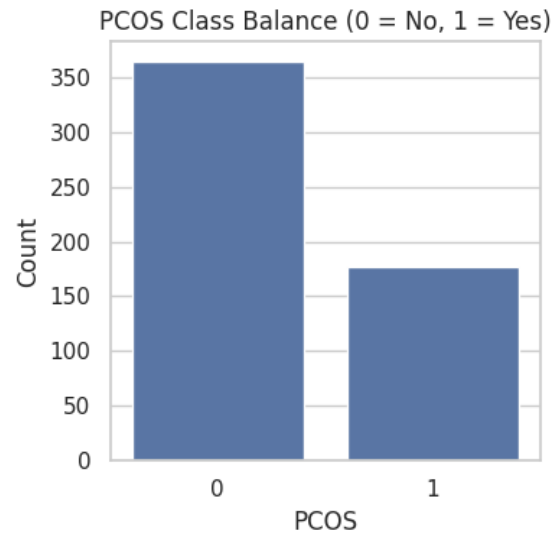


Figure 1. PCOS Class Balance (0 = No, 1 = Yes).

Missingness was reviewed across the merged dataset, with special attention to fields that had notable gaps, including Unnamed: 44, Marraige Status (Yrs), Fast food (Y/N), II beta-HCG(mIU/mL)_x, and AMH(ng/mL). Numerical features were imputed using median imputation to reduce sensitivity to outliers, while binary/flag-style variables (such as yes/no indicators) were filled using 0 when appropriate to represent the absence of a condition/response. Columns stored as object types were converted to numeric format (with coercion where needed) so the full feature matrix was compatible with machine learning models and scaling.

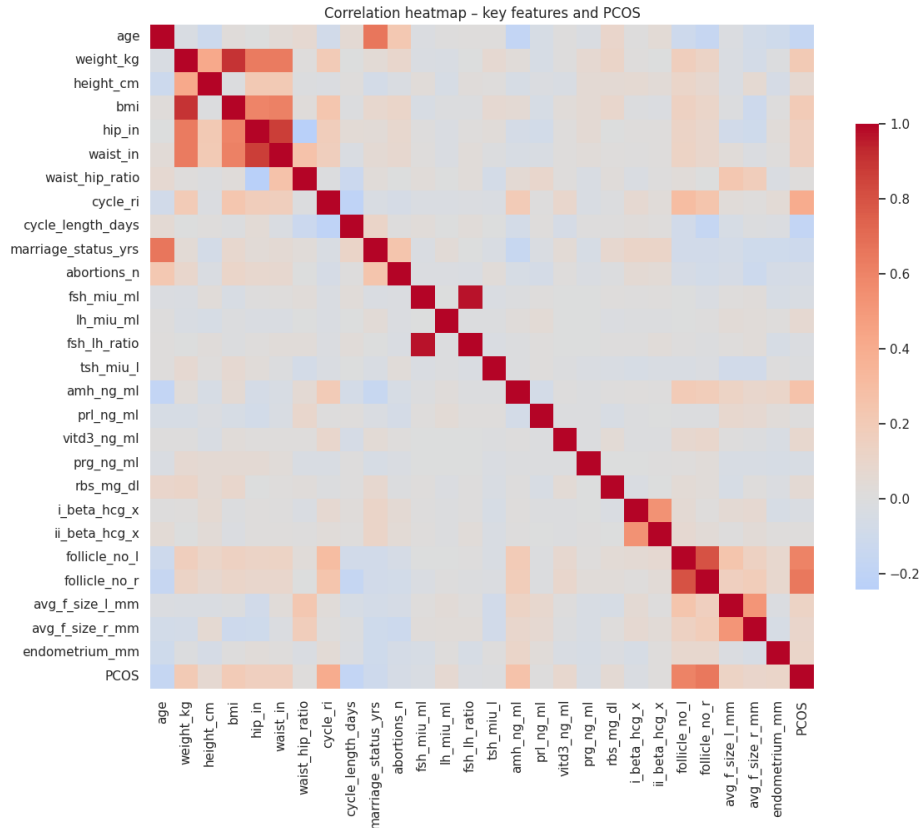


Figure 2. Correlation Analysis Clinical Dataset

As shown on the correlation analysis (Figure 2), there is strong multicollinearity among the body composition features. Weight_kg, height_cm, bmi, waist_in, and hip_in are all highly correlated with each other. For example, weight_kg has a very high correlation with bmi, waist_in, and hip_in. This means these variables are mostly capturing the same underlying information about body size and fat distribution. Keeping all of them could make our models less stable and harder to interpret.

The final dataset was split into training and testing sets using stratified sampling to preserve the PCOS vs No PCOS class ratio across splits. Numerical features were standardized using StandardScaler, fitted only on the training set and then applied to both training and test sets. This prevents data leakage and ensures fair evaluation, especially for models sensitive to feature scale (e.g., Logistic Regression and Neural Networks).

Image Data Preparation

To ensure data integrity and robust model training, a rigorous preprocessing pipeline was implemented for the unstructured ultrasound image dataset. The unstructured dataset comprises

11,700 ultrasound images, categorized into "Infected" (PCOS-positive) and "Non-infected" (Healthy) classes. Analysis of the class distribution indicates a relatively balanced dataset, with approximately 57% (n = 6,700) of images belonging to the Infected class and 43% (n = 5,000) to the Non-infected class. This balance is advantageous for deep learning applications, as it minimizes the bias often introduced by significant class imbalance, thereby reducing the necessity for synthetic oversampling techniques such as SMOTE.

Data Transformation and Normalization Pipeline: Prior to model ingestion, the raw unstructured ultrasound images underwent a standardized transformation pipeline to ensure numerical stability and dimensional uniformity. This process was implemented using the torchvision library.

Dimensional Resizing: All input images were resized to fixed spatial dimensions of 225 x 225 pixels. This step was necessary to ensure consistent tensor shapes (3 x 225 x 225) for the convolutional layers.

Tensor Conversion: The image data was converted from PIL images to floating-point tensors, scaling pixel intensity values to the range [0, 1].

Statistical Normalization: To prevent vanishing or exploding gradients and to accelerate convergence, we applied Z-score normalization. We explicitly calculated the channel-wise mean (μ) and standard deviation (σ) across the entire training subset. These statistics were then used to normalize the pixel values of the training, validation, and test sets according to the formula:

$$x_{norm} = \frac{x - \mu}{\sigma}$$

This ensures that the input features approximate a standard normal distribution centered at zero.

Models

Clinical Tabular Data - Classical Machine Learning Model

The following models were chosen for their distinct strengths and suitability for tabular classification tasks:

- **Logistic Regression (LR):** A simple yet powerful linear model, often serving as a strong baseline due to its interpretability and efficiency.

- Random Forest Classifier (RF): An ensemble method that builds multiple decision trees and merges their predictions to improve accuracy and control overfitting.
- Gradient Boosting Classifier (GB): Another ensemble method that builds trees sequentially, with each new tree correcting errors from previous ones, often leading to high predictive performance.
- XGBoost Classifier (XGB): An optimized distributed gradient boosting library designed for speed and performance, known for its robustness.

The dataset exhibited a class imbalance (fewer PCOS cases than non-PCOS). To prevent models from being biased towards the majority class, different techniques were applied:

- Logistic Regression & Random Forest: The `class_weight='balanced'` parameter was used. This automatically adjusts weights inversely proportional to class frequencies, giving more importance to the minority class (PCOS) during training.
- Gradient Boosting & XGBoost: Since these models do not typically have a `class_weight` parameter, a `sample_weight` array was constructed. The weight for the positive class (PCOS) was calculated as $\text{neg_count} / \text{pos_count}$ (approximately 2.06), meaning misclassifying a PCOS patient incurred a higher penalty.

Clinical Tabular Data - Neural Network

Multi-Layer Perceptron (MLP): A basic neural network architecture, chosen to explore the potential of deep learning on tabular data. The MLP consisted of several linear layers with batch normalization, ReLU activation, and dropout for regularization.

To handle class imbalance on the MLP model, a class-weighted cross-entropy loss function (`nn.CrossEntropyLoss`) was employed. The weights were calculated similarly to the gradient boosting approach ($w1 = \text{neg_count} / \text{pos_count}$), ensuring the model penalized misclassifications of the minority (PCOS) class more heavily.

Convolutional Neural Networks for Ultrasound Images

To address the binary classification problem of distinguishing PCOS-positive from PCOS-negative ovaries using ultrasound imagery, two complementary modeling paradigms were implemented:

1. A deep learning approach based on a custom Convolutional Neural Network (CNN) that learns hierarchical feature representations directly from image pixels.
2. A classical machine learning pipeline that combines hand-crafted visual descriptors, specifically Histogram of Oriented Gradients (HOG), with a Support Vector Machine (SVM) classifier.

The use of both paradigms serves two purposes. First, it enables a direct comparison between end-to-end feature learning and explicit feature engineering, which is of particular relevance in medical imaging, where interpretability and robustness are often prioritized alongside accuracy. Second, it provides a sanity check against overfitting or data leakage: a strong classical baseline reduces the likelihood that deep learning performance gains are purely artifacts of model capacity.

Convolution Neural Network: We architected a custom CNN designed specifically to capture the granular morphological features of ovarian follicles. The network processes a (Batch, 3, 225, 225) input tensor through three sequential convolutional blocks:

- Convolutional Block 1:
 - Layer: Conv2d (Input: 3 channels, Output: 16 channels, Kernel: 3x3, Padding: 1).
 - Activation: ReLU.
 - Pooling: MaxPool2d (Kernel: 2x2, Stride: 2).
 - Dimensionality Reduction: The spatial resolution is halved from 225x225 to 112x112.

This block performs initial edge and texture detection. The 3 x 3 filters capture local gradients, while the relatively small number of feature maps (16 channels) keeps computation tractable at the full image resolution.

- Convolutional Block 2:
 - Layer: Conv2d (Input: 16, Output: 32, Kernel: 3x3, Padding: 1).
 - Activation: ReLU.
 - Pooling: MaxPool2d (Kernel: 2x2, Stride: 2).
 - Dimensionality Reduction: Resolution decreases to 56x56.

As spatial resolution decreases, the number of feature channels doubles. This reflects a shift from low-level edge detection to more complex local structures, such as clusters of follicles or characteristic textural changes within the ovarian stroma.

- Convolutional Block 3:
 - Layer: Conv2d (Input: 32, Output: 64, Kernel: 3x3, Padding: 1).
 - Activation: ReLU.
 - Pooling: MaxPool2d (Kernel: 2x2, Stride: 2).
 - Dimensionality Reduction: Resolution decreases to 28x28.

The final convolutional block captures higher-order patterns and context, integrating information across larger receptive fields. The resulting feature maps encode rich morphological information in a compact representation.

Following these blocks, an AdaptiveAvgPool2d((1, 1)) layer is applied. This global pooling layer averages each 28 x 28 feature map (after the last pooling) into a single scalar, producing a 64-dimensional feature vector. Adaptive average pooling offers two practical advantages:

1. Size invariance: It decouples the final representation from the exact input resolution, making the architecture more robust to minor changes or future extensions in input size.
2. Regularization effect: By averaging over spatial locations, it encourages the model to focus on the presence of salient patterns rather than their exact positions, which is beneficial in ultrasound imaging where anatomical structures can shift within the frame.

- Classification Head:

To ensure the network is invariant to specific input sizes, we employed an AdaptiveAvgPool2d((1, 1)) layer, which collapses the spatial dimensions (64 x 28 x 28) into a flat vector (64 x 1 x 1). This feature vector is fed into a Multi-Layer Perceptron (MLP) consisting of:

- Linear Layer: 64 input features to 128 hidden units (ReLU activated).
- Output Layer: 128 hidden units to 2 output units (Logits for Infected/Non-infected).

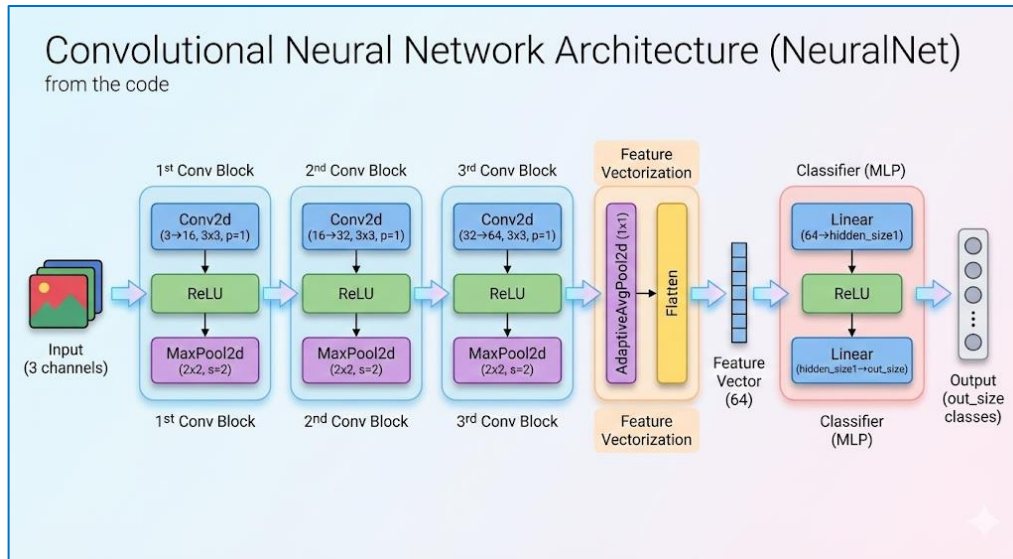


Figure 3: Convolutional Neural Network Architecture (for unstructured data)

Classical Machine Learning Approach - Feature Extraction for Unstructured Data

In parallel with the CNN, a classical computer vision pipeline was implemented to serve as a strong, interpretable baseline. This pipeline uses Histogram of Oriented Gradients (HOG) for feature extraction and a Support Vector Machine (SVM) for classification.

Rationale for HOG in Ultrasound Imaging:

HOG is a well-established descriptor that represents images in terms of local edge orientations. Instead of treating each pixel as an independent feature, HOG aggregates gradient information over small spatial cells and normalizes this information over larger blocks. This is advantageous in ultrasound imaging for several reasons:

- PCOS is characterized by distinctive spatial arrangements of follicles and changes in ovarian contour, both of which manifest as consistent edge patterns.
- Ultrasound images suffer from intensity fluctuations and speckle noise. HOG's block normalization improves robustness to such local illumination changes and noise.
- HOG is relatively low-complexity and interpretable, making it suitable for a baseline against more complex deep learning architectures.

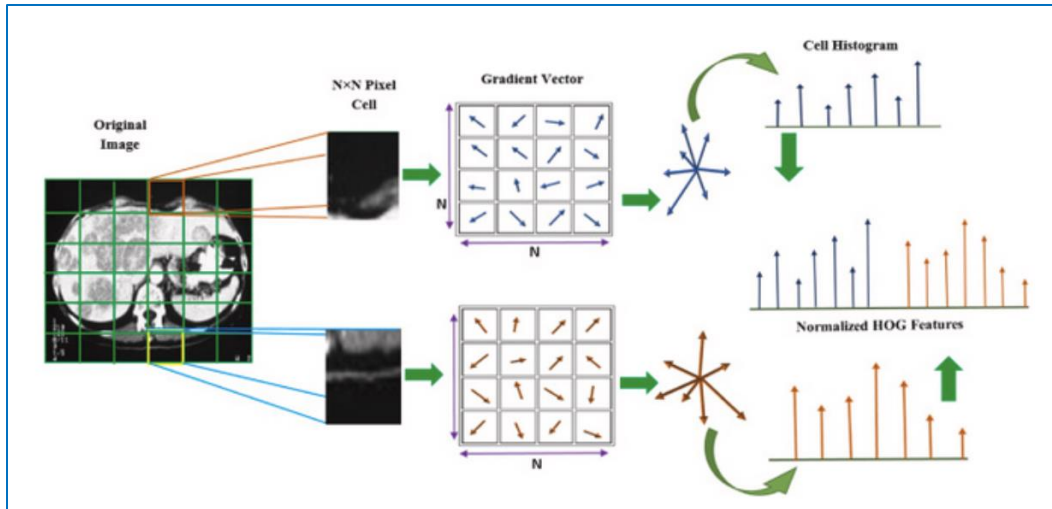


Figure 4: How Histogram of Oriented Gradients (HOG) works

Implementation Details:

HOG configuration:

HOG descriptors were computed using the following parameters:

- Orientations: 8

Gradient directions are quantized into 8 bins, enabling the descriptor to distinguish principal edge directions (horizontal, vertical, and diagonal). This granularity is sufficient to capture key anatomical boundaries without producing an excessively large feature space.

- Pixels per cell: (16,16)

Each cell covers a 16×16 region of the image. This cell size captures medium-scale texture and edge patterns, which is appropriate for representing follicle clusters and ovarian boundaries that span more than just a few pixels.

- Cells per block: (2,2)

Neighboring cells are grouped into blocks of 2×2 , within which the concatenated cell histograms are normalized. This block structure introduces local context and illumination invariance by comparing gradients across adjacent spatial regions.

The resulting HOG descriptors form a high-dimensional feature vector per image, encoding the distribution of gradient orientations across the entire ultrasound frame.

Dimensionality Reduction and SVM Classification:

Given the high dimensionality of HOG features, Principal Component Analysis (PCA) was employed as a dimensionality reduction step:

- PCA variance retention: 95%

Retaining 95% of the variance provides a compromise between preserving discriminative information and reducing noise and redundancy. This lower-dimensional representation improves computational efficiency and stabilizes subsequent SVM training.

The PCA-transformed features were finally fed into an **SVM classifier** for binary classification. Conceptually, the SVM seeks a decision boundary that maximizes the margin between the two classes in the transformed feature space. This margin maximization principle often yields strong generalization performance, particularly when the data are well represented by the underlying feature space (here, the combination of HOG and PCA).

Hyperparameter Tuning and Training Procedure

Hyperparameter Tuning Clinical Tabular Data

Hyperparameter tuning was performed in three rounds, with each round using the previous results to refine the next set of choices. The main goal was to improve performance in a clinically meaningful way, with emphasis on ROC–AUC and Recall for the positive class (PCOS = 1). This approach avoids selecting a model that looks strong on overall accuracy but underperforms on identifying PCOS cases.

Round 1 (initial tuning). Logistic Regression was updated by switching the solver to `saga`, since it is well-suited for larger datasets and supports multiple regularization options. For Random Forest, `max_depth = 10` was introduced to limit tree growth and reduce overfitting, and `max_features = 'log2'` was used so each split considers a smaller subset of features, which can improve generalization. For both Gradient Boosting and XGBoost, a conservative learning strategy was tested using `learning_rate = 1e-3` combined with `max_depth = 10`, exploring a more complex tree structure while learning in very small steps.

Round 2 (refinement). Based on Round 1 patterns, Logistic Regression kept `solver = 'saga'` since it remained stable and competitive. Random Forest increased `max_depth` to 50 to allow the model to capture more complex relationships, while keeping `max_features = 'log2'` as a

control against overfitting. For Gradient Boosting and XGBoost, the learning rate was increased to 0.05 and tree depth reduced to 5. The earlier learning rate was likely too small and risked underfitting; this adjustment aimed to improve learning efficiency while controlling model complexity.

Round 3 (final refinement). Round 3 focused on stability. Logistic Regression remained unchanged due to consistent performance. Random Forest maintained `max_depth = 50`, and both Gradient Boosting and XGBoost kept `learning_rate = 0.05` and `max_depth = 5`, as these settings provided strong performance without introducing unnecessary complexity.

A Multi-Layer Perceptron (MLP) was also tuned to evaluate a deep learning baseline for tabular data. The network used an input dimension matching the feature set (43 features), two hidden layers with 64 units, and an output size of 2 for binary classification. Batch normalization (`BatchNorm1d`) and ReLU activations were applied in the hidden layers, with Dropout (0.3) after each activation to reduce overfitting. To address class imbalance, the model was trained using `CrossEntropyLoss` with class weights. Optimization used Adam with a learning rate of $1e-3$. Training initially ran for 50 epochs and was extended to 200 epochs during tuning to allow the network additional time to converge.

Hyperparameter Tuning Ultrasound Images

- Same subsection covers tuning for both DL and classic ML models.
- Classification metrics: accuracy, precision, recall, F1, AUC, confusion matrices, ROC curves.
- If you end up with any regression task, add MSE/MAE here.
- Explain which metrics you prioritize and why.

Evaluation Metrics

This comprehensive evaluation provides a strong foundation for understanding the strengths and weaknesses of each model in predicting PCOS.

Tabular Dataset Results

Classical ML + Neural Network Performanc

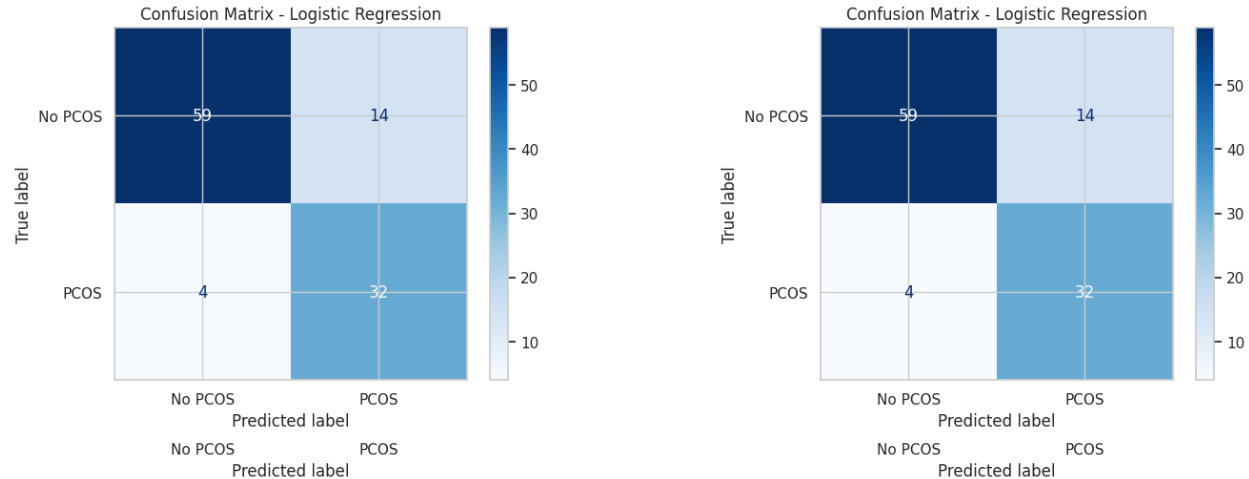


Figure 5. Baseline ML models confusion matrix

The confusion matrix makes the model's behavior very clear by showing exactly where errors occur. In this result, the model correctly classified 59 patients as No PCOS (true negatives) and 32 patients as PCOS (true positives). At the same time, it produced 14 false positives (patients without PCOS predicted as PCOS) and 4 false negatives (patients with PCOS predicted as No PCOS). This is important because it shows performance beyond accuracy: the model is not just “getting a good score,” it is specifically making relatively few mistakes on the PCOS class.

From a screening perspective, the most critical value is the false negatives (FN = 4), since these represent missed PCOS cases that could delay follow-up or treatment. The low FN count supports a high PCOS recall, meaning the model is sensitive to identifying positive cases. However, the false positives (FP = 14) show the trade-off: the model tends to flag some No PCOS patients as PCOS, which can lead to extra follow-up testing and potential patient anxiety. Overall, this confusion matrix suggests the model leans toward catching more PCOS cases, accepting some additional false alarms to avoid missing true positives.

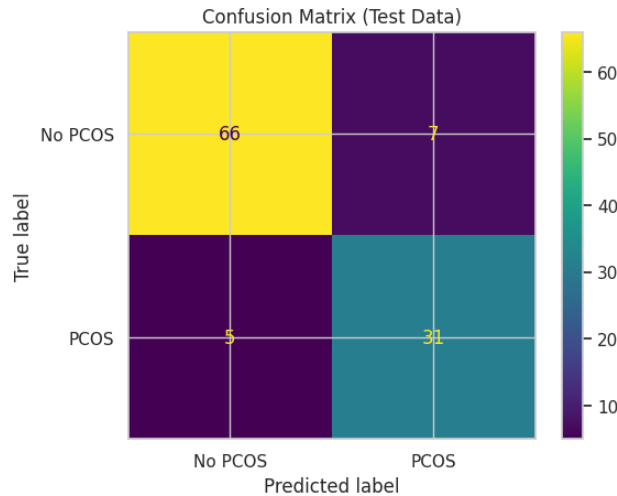


Figure 6. Baseline MLP model confusion matrix

The PyTorch MLP model demonstrates strong predictive performance for PCOS classification, particularly evidenced by an ROC AUC of (0.9521) and an overall accuracy of approximately (88%) on the test set. The model's ability to classify both 'No PCOS' and 'PCOS' cases with good precision and recall indicates its potential for real-world application.

The confusion matrix shows that the model correctly identifies most PCOS cases while keeping a relatively low number of false positives. The combination of class weighted loss and the MLP architecture gave a good balance between sensitivity to PCOS and overall performance, which is important for this type of medical screening task.

	PCOS_Precision	PCOS_Recall	PCOS_F1-Score	No_PCOS_Precision	No_PCOS_Recall	No_PCOS_F1-Score
Logistic Regression (Baseline)	0.6957	0.8889	0.7805	0.9365	0.8082	0.8676
Random Forest (Baseline)	0.9310	0.7500	0.8308	0.8875	0.9726	0.9281
Gradient Boosting (Baseline)	0.9375	0.8333	0.8824	0.9221	0.9726	0.9467
XGBoost (Baseline)	0.8824	0.8333	0.8571	0.9200	0.9452	0.9324
MLP	0.8000	0.8889	0.8421	0.9420	0.8904	0.9155

Table 1. Per-Class Metrics Comparison (Baseline Classic ML and MLP)

Tuned Models

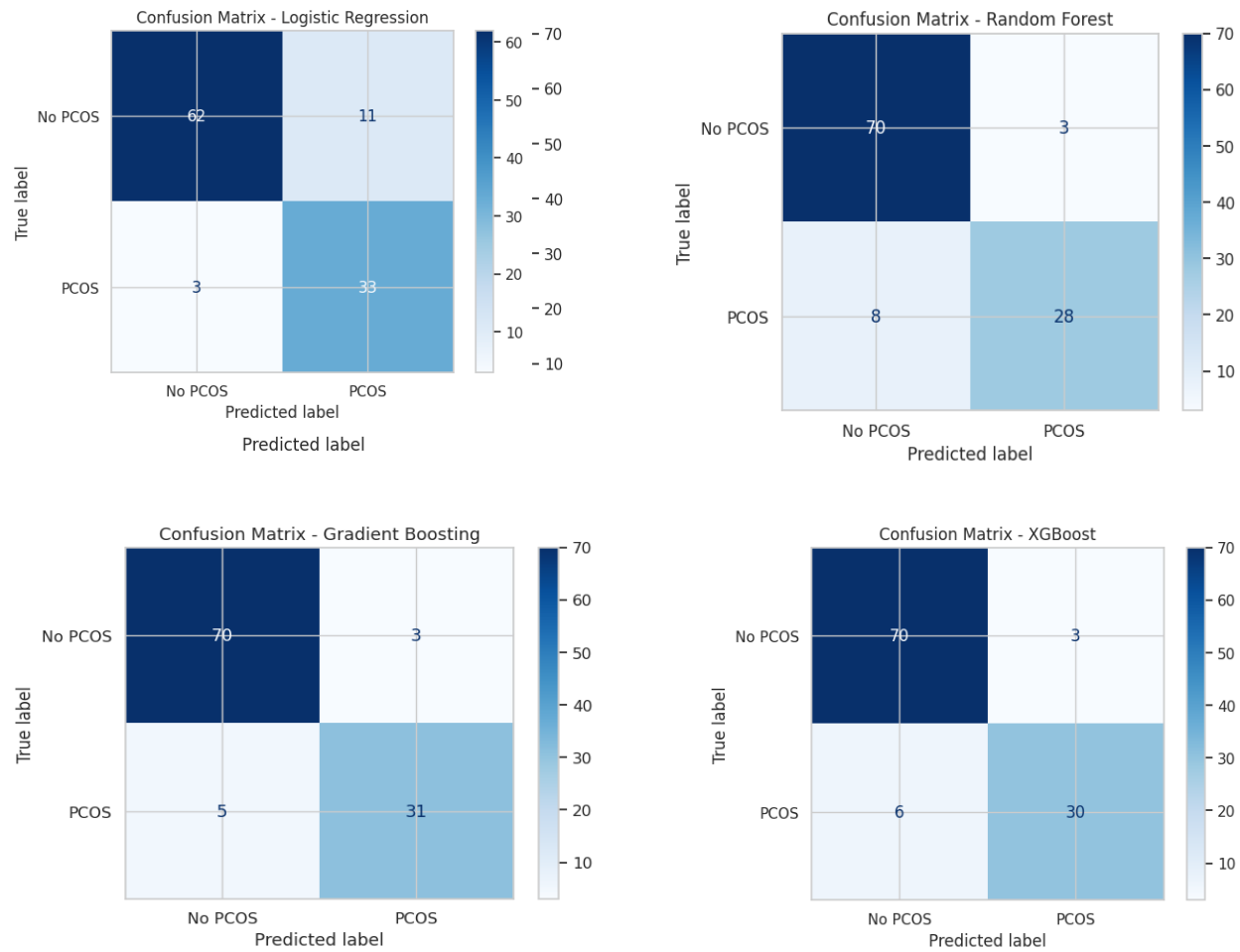


Figure 7. Tuned ML Models Confusion

Matrix

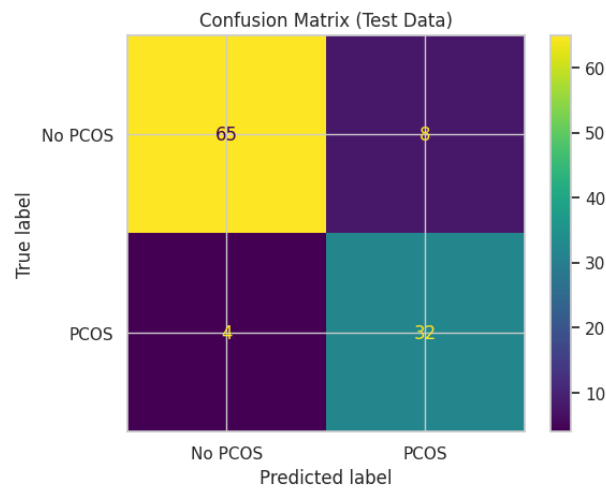


Figure 8. MLP model confusion matrix

Hyperparameter Tuning Impact: Iterative tuning refined parameters like learning_rate, max_depth, and solver for classic ML models, and increased MLP training to 200 epochs, generally leading to slight performance improvements over baseline models.

Model	No PCOS_Precision	No PCOS_Recall	No PCOS_F1-Score	PCOS_Precision	PCOS_Recall	PCOS_F1-Score
Logistic Regression	0.9538	0.8493	0.8986	0.7500	0.9167	0.8250
Random Forest	0.8974	0.9589	0.9272	0.9032	0.7778	0.8358
MLP	0.9420	0.8904	0.9155	0.8000	0.8889	0.8421
XGBoost	0.9211	0.9589	0.9396	0.9091	0.8333	0.8696
Gradient Boosting	0.9333	0.9589	0.9459	0.9118	0.8611	0.8857

Table 2. Per-Class Metrics Comparison (Tuned Classic ML and MLP)

The recall and F1-score comparison plots make the minority-class story (PCOS) much clearer than accuracy alone. For PCOS recall, the tuned Logistic Regression model stands out as the strongest option (0.9167), meaning it identifies the highest proportion of true PCOS cases and minimizes false negatives. The MLP is also consistently strong on recall (0.8889), which suggests a neural network can perform well on this tabular feature set without needing a complex architecture. It is also notable that baseline Logistic Regression already had high recall, which indicates this model family is naturally sensitive for this dataset and tuning mainly sharpened that strength.

When switching focus to F1-score for PCOS, the top performers shift slightly. Tuned Gradient Boosting and XGBoost achieve the best PCOS F1-scores (≈ 0.8857 and ≈ 0.8696), showing a more balanced trade-off between precision and recall for the positive class. This highlights the key clinical trade-off: Logistic Regression (tuned) maximizes recall but does so with lower PCOS precision than the best ensembles, which can reduce its F1-score. In practical terms, if the priority is not missing PCOS cases (screening mindset), recall is the deciding factor. If false positives create substantial cost, burden, or anxiety and the goal is a steadier balance, PCOS F1-score becomes more appropriate for selecting the model.

For the No PCOS class, the plots show consistently high recall and F1 across nearly all models (generally >0.90). This is expected given the class imbalance: most models learn the majority class well. The tuned ensemble models especially Random Forest, Gradient Boosting, and XGBoost, remain strong, with very high No PCOS recall (0.95+) and F1 (0.92+), meaning they are reliable at correctly identifying patients who do not have PCOS. This matters because it

reduces unnecessary flags, but it is not usually the limiting factor since performance on the majority class is already strong across the board.

Overall, the tuning results show meaningful improvements where it matters most: the PCOS class. Logistic Regression benefits the most from tuning, improving both PCOS recall and its overall balance, reinforcing the value of hyperparameter optimization. For Random Forest, Gradient Boosting, and XGBoost, tuning tends to provide smaller gains, mostly improving the balance of PCOS precision/recall rather than drastically changing rankings. Taken together, the plots support a clear model-selection framing: Logistic Regression (tuned) (and MLP) are strongest when the main goal is maximizing PCOS detection, while Gradient Boosting (tuned) and XGBoost (tuned) are best when the goal is balanced PCOS performance with fewer false positives.

Ultrasound Image Dataset Results

CNN Performance

All final models were evaluated on a **held-out, stratified test set** comprising **2,358 ultrasound images**. Stratification ensured that the class proportions in the test set mirrored the overall distribution (Infected vs. Non-infected), thereby providing an unbiased estimate of generalization performance.

The following metrics were computed:

- Accuracy: Overall proportion of correctly classified images.
- Precision: For each class, the proportion of predicted positives that are true positives.
- Recall (Sensitivity): For each class, the proportion of true positives that are correctly identified.
- ROC–AUC: Area under the Receiver Operating Characteristic curve, summarizing the trade-off between sensitivity and specificity across decision thresholds.

Baseline Model:

- CNN Baseline (LR 1×10^{-3} , No Dropout)

Class/Metric	Precision	Recall	F1-Score	Support
0	0.96	0.97	0.97	4074

1	0.96	0.95	0.95	2996
accuracy			0.96	7070
macro avg	0.96	0.96	0.96	7070
weighted avg	0.96	0.96	0.96	7070

Table 3. Baseline CNN Accuracy Training Output

Class/Metric (Test Dataset)	Precision	Recall	F1-Score	Support
0	0.96	0.97	0.97	1357
1	0.96	0.95	0.95	1001
accuracy			0.96	2358
macro avg	0.96	0.96	0.96	2358
weighted avg	0.96	0.96	0.96	2358

Table 4. Baseline CNN Accuracy Test Output

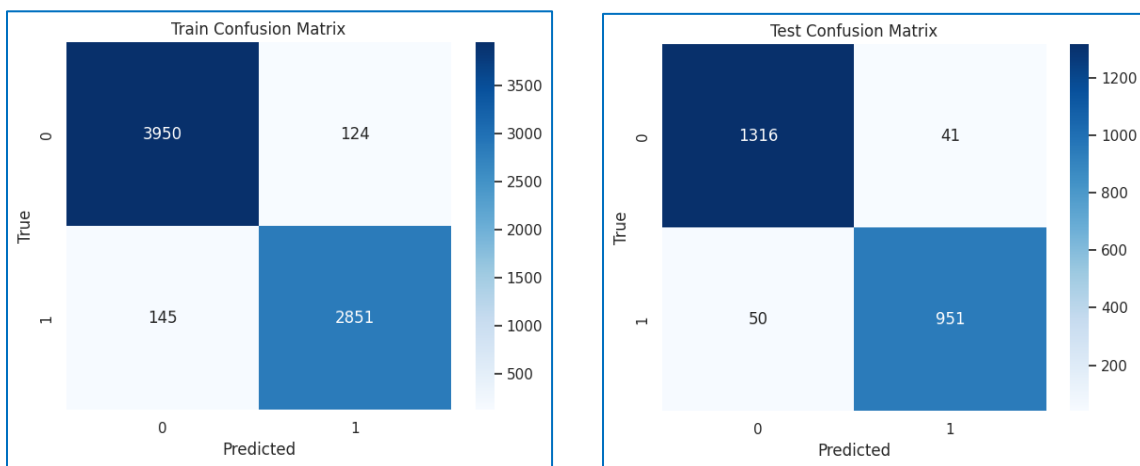


Figure 9. Train and Test Confusion Matri

The Baseline CNN achieves the highest overall accuracy (96%) and demonstrates balanced performance across both classes. This balance is essential in medical applications, where prioritizing one class at the expense of the other (e.g., maximizing sensitivity while severely compromising specificity) can lead to clinical consequences such as unnecessary interventions or missed diagnoses.

Analysis of the confusion matrix for the Baseline CNN reveals:

- False Negatives (Infected misclassified as Non-infected): 50

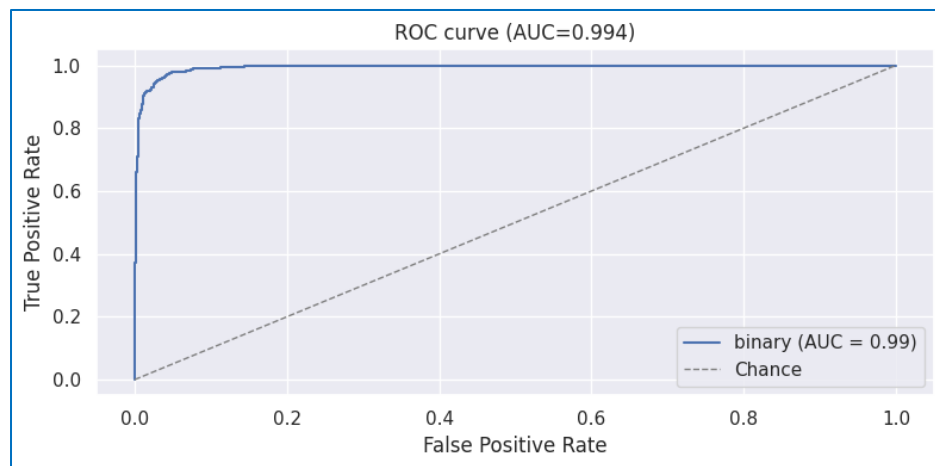
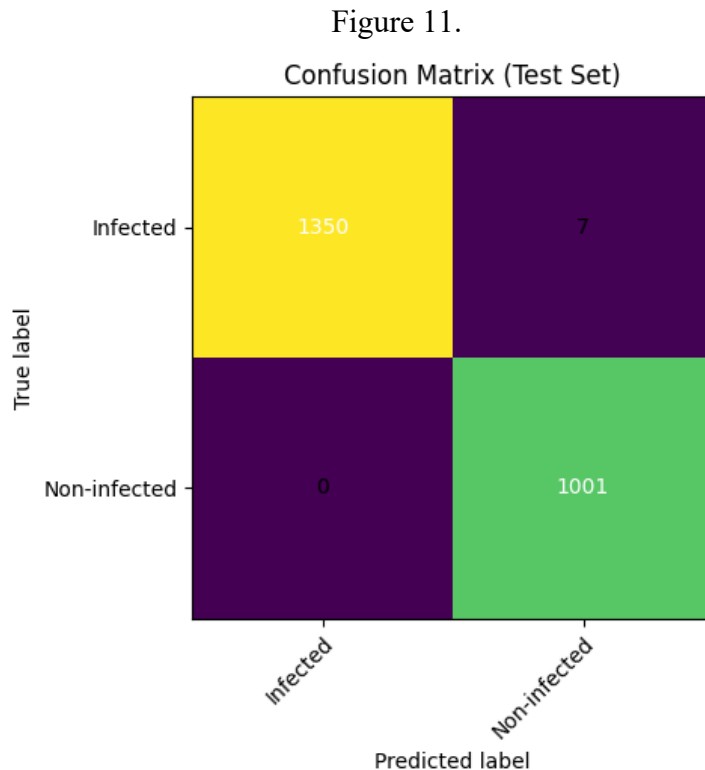


Figure 10.

- False Positives (Non-infected misclassified as Infected): 41

From a clinical perspective, False Negatives are particularly critical, as they correspond to patients with PCOS who would not be flagged by the system and might fail to receive timely follow-up. The relatively low number of False Negatives indicates strong sensitivity. At the same

time, the limited number of False Positives suggests that the model does not excessively overpredict the Infected class.



The ROC curve for the Baseline CNN yields an Area Under the Curve (AUC) of 0.994, a value that is very close to the theoretical maximum of 1.0. This indicates that, across a wide range of decision thresholds, the model maintains excellent separability between Infected and Non-infected cases.

Classical ML on Extracted Features (Without Neural Networks)

For the HOG + SVM model, performance is primarily governed by:

1. The effective dimensionality of the feature space (controlled via PCA).
2. The rigidity of the decision boundary (controlled via the SVM regularization parameter

A grid search was conducted over the following hyperparameters:

- PCA variance retention: {90%, 95%, 99%}
- SVM regularization parameter C: {10,100}

The configuration with 95% variance retention and appropriately tuned C yielded the best overall trade-off between complexity and generalization. Under this configuration, the HOG + SVM model achieved:

- **Validation accuracy:** 0.9953 (99.53%)
- **Test accuracy:** 0.9958 (99.58%)

Class/Metric (Test dataset)	Precision	Recall	F1-Score	Support
infected	1.00	0.99	1.00	1357
noninfected	0.99	1.00	1.00	1001
accuracy			1.00	2358
macro avg	1.00	1.00	1.00	2358
weighted avg	1.00	1.00	1.00	2358

Table 5.

The validation classification report further highlights this strong performance:

- Validation precision:
 - Infected: 1.00
 - Non-infected: 0.99
- Validation recall:
 - Infected: 0.99
 - Non-infected: 1.00

These results indicate a nearly perfect separation between classes in the HOG + PCA feature space, suggesting that the hand-crafted features are exceptionally well aligned with the underlying structure of the data.

Best Models and Confusion Matrices / ROC Curves:

- CNN Regularized (LR 3×10^{-4} , Dropout 0.3)

Class/Metric (Train Dataset)	Precision	Recall	F1-Score	Support
0	0.96	0.93	0.94	4074
1	0.91	0.94	0.92	2996
accuracy			0.93	7070
macro avg	0.93	0.94	0.93	7070
weighted avg	0.94	0.93	0.93	7070

Table 6.

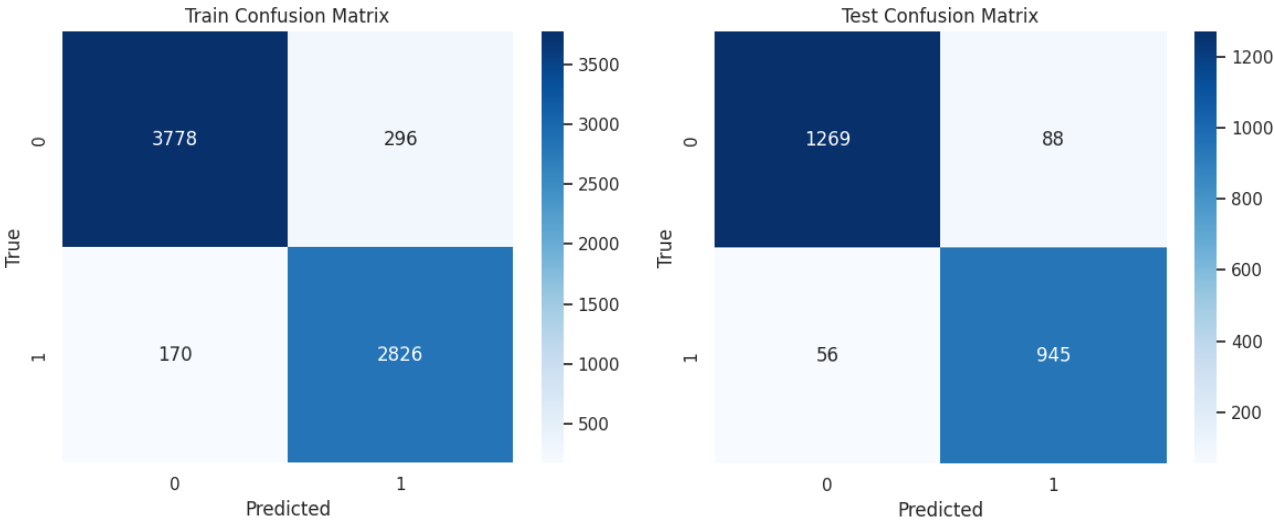


Figure 12.

Class/Metric (Test Dataset)	Precision	Recall	F1-Score	Support
0	0.96	0.94	0.95	1357
1	0.91	0.94	0.93	1001
accuracy			0.94	2358

macro avg	0.94	0.94	0.94	2358
weighted avg	0.94	0.94	0.94	2358

Table 7.

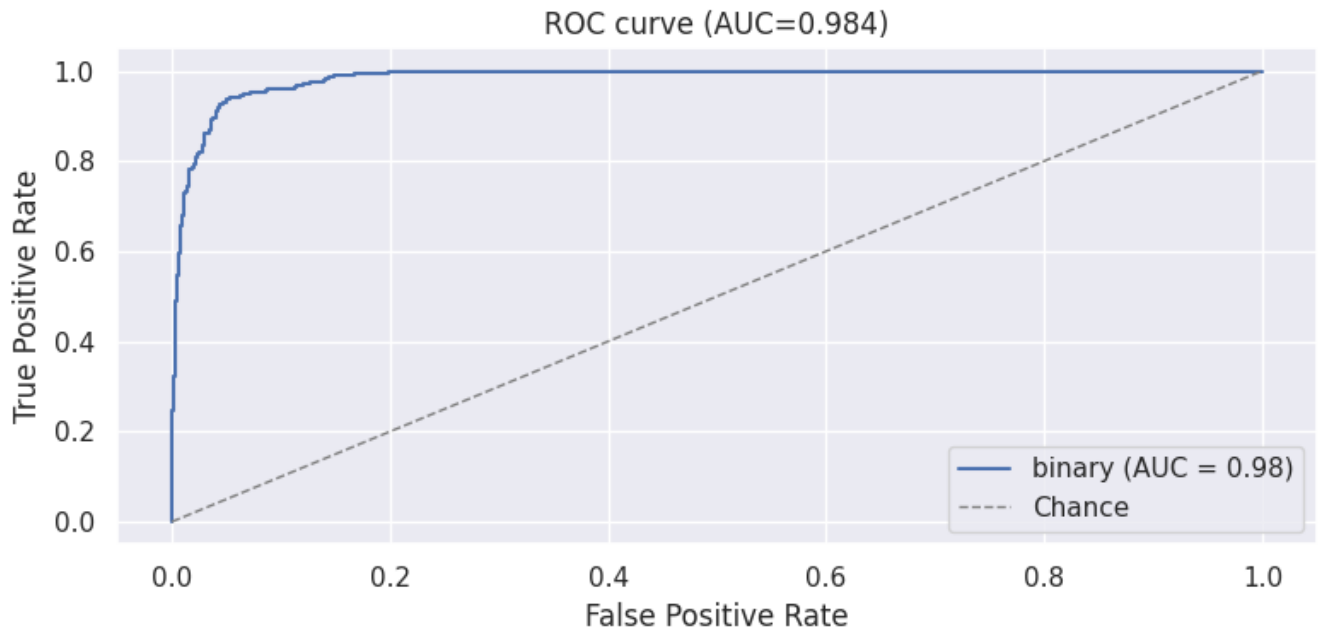


Figure 13.

A central methodological finding is the comparative behavior of the Baseline and Regularized CNN models. Conventional wisdom suggests that adding regularization, such as Dropout, should typically improve generalization performance, especially on limited datasets. However, in this study:

- The Baseline CNN without Dropout outperformed the Regularized CNN by 2 percentage points in test accuracy (96% vs. 94%).
- Training and validation accuracies for the Baseline model were closely aligned (94.9% vs. 96.4%), indicating that substantial overfitting was not present.

Several plausible explanations can be considered:

1. High Signal-to-Noise Ratio in the Feature Space:

The ultrasound dataset appears to contain distinctive and relatively consistent morphological signatures of PCOS. When the underlying signal is strong, the model may generalize well without aggressive regularization.

2. Architectural Implicit Regularization:

The use of global average pooling and a relatively shallow network implicitly constrains the model's capacity. This can act as a form of built-in regularization, reducing the need for additional mechanisms such as Dropout.

3. Underutilization of Capacity in the Regularized Model:

Introducing Dropout may have prevented the model from fully utilizing the 128-dimensional hidden layer, thereby limiting the richness of the learned representation. In this scenario, the Regularized CNN might be slightly underfitting relative to the complexity of the task.

This result underscores an important practical message: more regularization is not always better. However, in this case we can see that the baseline is aggressive and accurate; regularized model is more robust with unseen data noise. Optimal regularization is data and architecture-dependent and should be empirically validated rather than assumed.

Discussion

Results and Implications

Given the very nature of the dataset selected for this project it is not surprising that the most viable model would prove to be the traditional (even to the point of baseline) CNN model. It is for this very purpose that such neural networks exist! The way in which these follicles present on an ultrasound, creating voids or pockets of black space within the ultrasound create a solid threshold for training and classification without the need for manual feature adjustments, at least for the purposes of this application.

In medical cases one could use this to flag potential issues for investigation, but there are of course many different medical conditions that could present with the same sorts of imaging features as PCOS on an ultrasound. It would be both unwise and medically unsafe to claim that

the sanitized and pre-prepared dataset used in the analysis of this project is capable of anything other than a ‘proof of concept.’

On the topic of the other models in this project, while it is also safe to say that the utilization of HOG for the aggregation of gradient data and vector selection has its use cases, it is not a viable format for building a medical model. However, there is also substantial proof that while the traditional Machine learning models for the tabular data may have lacked the more sophisticated modeling capabilities of the CNN, they are still able to deliver quality results that are at least comparable to that of the CNN model presented.

As was predicted at the beginning of the project, tree-based methods proved to be the most effective and competitive, combined with gradient boosting it surpassed the MLP deep learning model. This is also an unfair characterization of the MLP model for several reasons, primarily due to the small size of the data analysis as well as the limited number of features the model was trained on.

Because the dataset presented for presentation in this project was made publicly available through Kaggle, we had the ability to check the veracity of our model's accuracies and conclusions based on the works of other individuals. While model sophistication varied greatly across these other publicly available projects published on Kaggle, when looking at a plurality of the notebooks presented using the [PCOS Detection with Ultrasound images](#) dataset, most notebooks demonstrated a similar accuracy of ~98-99% accuracy through various methods of multilayer CNNs. This stands to reason as the goal of medical classification is to increase the accuracy of the model as far as reasonably possible without compromising stability.

There is more variation given that the number of different features and models can be utilized in the [Polycystic ovary syndrome \(PCOS\)](#) tabular dataset. Still, most individuals utilized a similar approach to the models presented in this project and again posted similar results, with accuracy for tree-based models ranging from ~88%-94% accuracy. The example below is provided by user [İlayda Duratnir](#) in her model titled: PCOS Classification, where she ran a similar suite of models resulting in accuracies within a near margin of our presented models. In

fact, the model accuracy of our XGBoost model of .917431 while her model's accuracy was .9127.

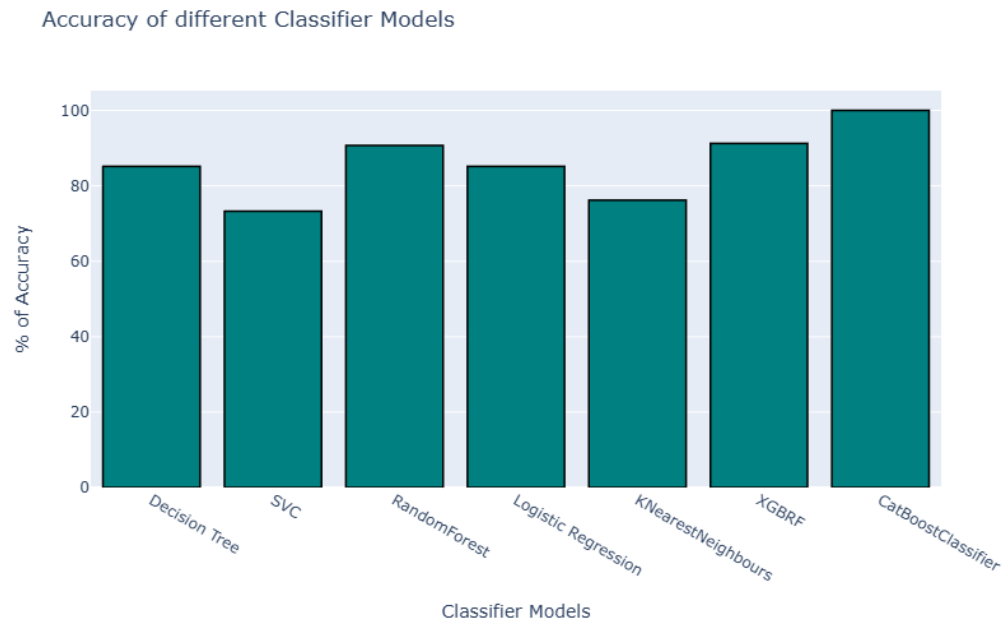


Figure 14. Accuracy of Classifier Models from [İlayda Duratnir](#)

There is still potential for the machine learning model to function as a model with equal practicality to the CNN, however. Other academic works frequently will utilize multimodal analysis to boost the accuracy of the overall model. In the case of the tabular dataset used to train this model, there would need to be an in-depth analysis of the embedded bias within the clinical features before the model could be retrained. Features in the dataset such as BMI, weight, cycle regularity, hirsutism scores, and metabolic markers overlap heavily with protected characteristics.

Current Research Landscape

While preparing for this presentation, there were two distinct studies that stood out based on their similarities and differences to the approaches that we have taken within the bounds of our own minor attempts at model creation. The first was Suha and Islam’s “Extended Machine Learning Technique for PCOS Detection Using Ovary Ultrasound Image”, and the second was Hosain et al. “PCONet: A Convolutional Neural Network Architecture to Detect PCOS”. What made Suha and Islam’s model interesting was both the reliance on multimodal methodology for capturing the highest level of accuracy while integrating the complete suite of features that we

originally hoped to capture, but proved beyond our current abilities, but also the focus on the layers of image classification and analysis including dynamics such as geometry, location as well as image optimization techniques. Naturally even the machine learning models presented in the study boasted ~99% accuracy, but with a wide breadth of analysis to demonstrate the veracity of the model accuracy.

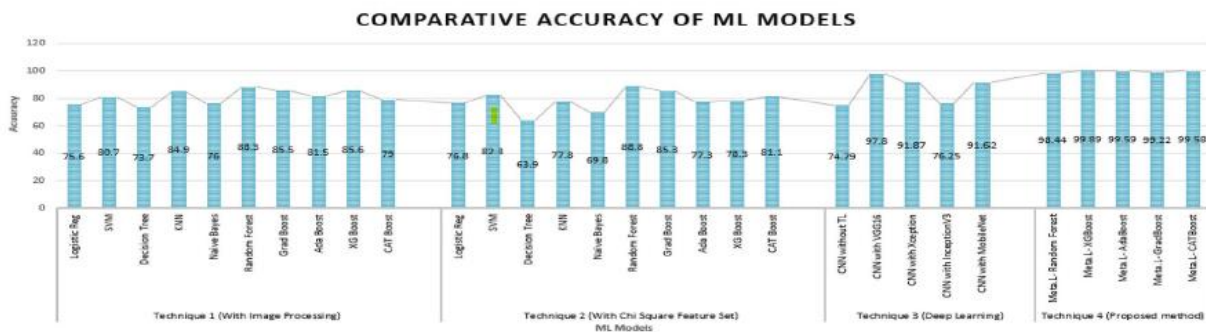
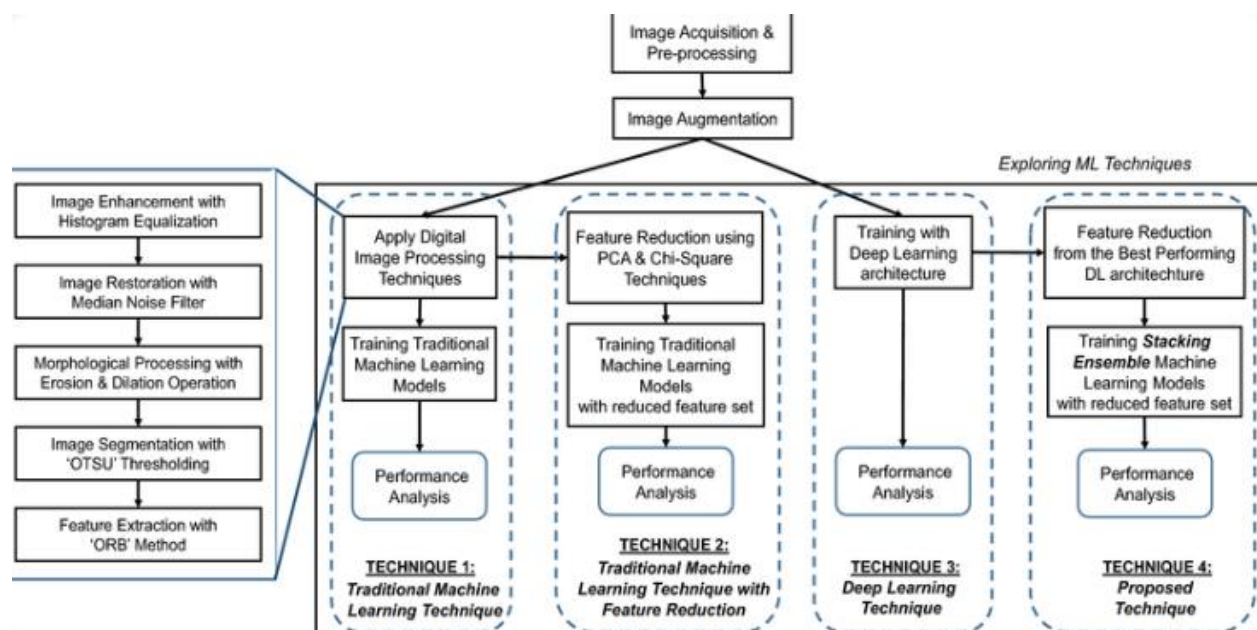


Figure 15. Comparative ML Accuracies from Sussha & Islam's "Extened Machine Learning"

In the image below we are also able to look at the sequencing and types of techniques that could be implemented to further improve our own model as presented, as well as the type of digital preprocessing that was utilized to tackle the same issues with transforming image data into a parable machine learning format, as noted in the callout for the 'Apply Digital Image Processing Techniques' step post image augmentation under 'Technique 1'.

Figure 16. Multimodal Training Model from Sussha & Islam's "Extened Machine Learning"



Alternately, the important takeaway from the PCOSNet article by Hosain, Mehedi and Kabir was the arbitrary limitations we placed regarding the depth and complexity of our CNN. Given the similar nature of the dataset utilized within the study as well as our tools, there should have been a push to add additional convolutional layers beyond the two included in the final model. Below we can see the layout of the bespoke PCOSNet model implemented in the study next to the confusion matrix for the test set.

TABLE I: Outline of PCONet

Type	Specifications	Output Size
Conv_1	32(3, 3), s=1	(222, 222, 32)
Max_pool_1	(2, 2), s=2	(111, 111, 32)
Conv_2	32(3, 3), s=1	(109, 109, 32)
Max_pool_2	(2, 2), s=2	(54, 54, 32)
Conv_3	64(3, 3), s=1	(52, 52, 64)
Max_pool_3	(2, 2), s=2	(26, 26, 64)
Conv_4	64(3, 3), s=1	(24, 24, 64)
Max_pool_4	(2, 2), s=2	(12, 12, 64)
Conv_5	128(3, 3), s=1	(10, 10, 128)
Max_pool_5	(2, 2), s=2	(5, 5, 128)
Flattening Layer		(None, 3200)
Dense Layer 1	128, ReLu	(None, 128)
Dense Layer 2	256, ReLu	(None, 256)
Output Layer	2, Sigmoid	(None, 2)

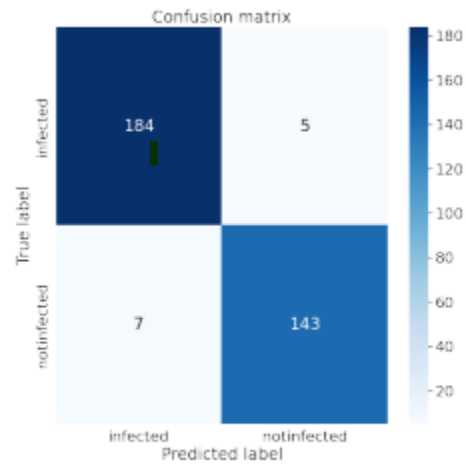


Fig. 14: Confusion matrix of PCONet on test set.

Figure 17. PCONet CNN Model & Confusion Matrix

This of course begs the question that even with utilizing the considerable power of Kaggle's freely (but time limited) available GPU T4x2, would we possess the necessary processing power and time to be able to run the data through as complex a model. While there is a difference in scale between the dataset used in the study and the Kaggle dataset, the overall results gained were similar to our best running CNN:

TABLE II: Comparative table of performance evaluation parameters conducted on test set for both models.

Parameters	Class	InceptionV3	PCONet
Accuracy		96.56%	98.12%
Precision	Cystic	0.94	0.96
	Healthy	1.00	0.97
Recall	Cystic	1.00	0.97
	Healthy	0.91	0.95
F1 Score	Cystic	0.97	0.97
	Healthy	0.95	0.96

Figure 18. Comparison between PCONet & Project CNN Accuracies

It should be stated that there is a considerable difference in scale between the test data in the

```
Confusion matrix on TEST:
[[1277  80]
 [   9 992]]
Classification report on TEST:
              precision    recall  f1-score   support

     0       0.99      0.94      0.97      1357
     1       0.93      0.99      0.96      1001

 accuracy          0.96      2358
 macro avg       0.96      0.97      0.96      2358
 weighted avg    0.96      0.96      0.96      2358
```

study and the Kaggle dataset. Overall number of test cases for the PCOSNet was 339 total to our 2358. The data size brings up a number of pertinent questions that get to the heart of the issues surrounding the use of CNN's and the identification of PCOS in a medical setting.

Business Use Analysis

Given the clear ability of established studies to identify the presence of PCOS in women as well as the prevalence of the condition itself within the population of ovary possessing people: why do we not see a clear implementation of these techniques within the medical field? The answer to this is a commonality shared between both published studies as well as this very project. The data that is gathered to produce these models is academic, not clinical. Setting aside the models need to identify solely instances of PCOS within the ovaries, and no other anomalies, the diagnostic complexity of properly identifying and diagnosing PCOS does not allow for the creation of a clear, consolidated set of guidelines that allow for the creation of a broadly inclusive set of features in which models can be trained on. There are currently three accepted diagnostic guidelines for PCOS: NIH 1990, Rotterdam 2003, and AE-PCOS 2006 frameworks. Each of these presents sometimes conflicting sets of features that would be required to even being the consolidation of a usable dataset.

Even after the benchmark of diagnostic features is settled, there then becomes an issue with building the proper scale of data given the populations such a model would be deployed in. While there is a great deal of data that can be collected from the general wellness visits of individuals across the world, most of them will be of a socio-economic status that grants them access to the level of care that allows for the collection of ultrasound data, creating embedded pockets of bias within any collected data. The ability to identify a usable control group becomes nearly impossible given the breath of the population a proper model would be deployed to benefit.

Next Steps

Despite these hinderances to successful deployment of medically trained models, the continued creation of these academic level models is not without value. Data can still be aggregated and using a variety of techniques for bias mitigation and feature engineering there is

still an opportunity to see such instances of multimodal CNN's implemented in medical settings. There are many implicit benefits to this as well, considering that 70% of the global population of individuals with ovaries will miss diagnosis.

The medical infrastructure that supports the scanning for PCOS is widely available and reasonably accessible, even to rural or economically disadvantaged areas of the globe.

Increasing positive health outcomes also has a rising effect on fetal and infant health as there are many additional side effects that can cause harm to both the mother and child, including maternal morbidity, miscarriage and gestational diabetes.

References

- Hosain, A, et al. *PCONet: A Convolutional Neural Network Architecture to Detect Polycystic Ovary Syndrome (PCOS) from Ovarian Ultrasound Images*. 2022.
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